

A Generated Split-Forest Decidability Proof for $\mathcal{ALCI}_{\text{RCC5}}$ Without Automata: A Repaired Containment-Oriented Presentation

Prepared as a consolidated proof manuscript

May 21, 2026

Abstract

We give a self-contained repaired proof of decidability for concept satisfiability in $\mathcal{ALCI}_{\text{RCC5}}$ over strong abstract atomic RCC5 frames. The proof keeps the original split-forest architecture: a containment-oriented generated backbone for PP/PPI, split copies for multiple containment parents, finite sibling interfaces, and a typed equality quotient. The proof removes three sources of unsoundness. First, the vertical cover relation is a generated witness-cover relation, not the Hasse diagram of an arbitrary semantic PP-order. Second, blocking repetition is separated from semantic equality. Third, upward PP eventualities are discharged through equality-mate occurrences, so an element may have several incomparable selected superpart witnesses. No Buchi or parity automata are used. Regularity is obtained by a finite-index blocking lemma with explicit request-fulfilled cycles, and the final decision procedure enumerates finite split-forest certificates and checks only finite local, reachability, equality-congruence, and mosaic conditions.

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1 Statement of the theorem

The input is a concept C_0 in negation normal form over concept names and the five RCC5 atoms used as modal roles. The role set is

$$B = \{\text{EQ}, \text{DR}, \text{PO}, \text{PP}, \text{PPI}\}.$$

The intended result is the following.

Theorem 1.1 (Main decidability theorem). *Concept satisfiability for $\mathcal{ALCI}_{\text{RCC5}}$ over strong abstract complete atomic RCC5 frames is decidable. Equivalently, for every input concept C_0 ,*

$$C_0 \text{ is satisfiable} \iff C_0 \text{ has a valid finite generated split-forest certificate.}$$

The right-hand side is effectively enumerable and decidable by finite checks.

The proof has the following shape. First, every arbitrary satisfying abstract model can be thinned to a witness-generated induced submodel. This justifies working only with sparse generated witness-cover structures. Second, every witness-generated model has a split occurrence presentation in which vertical generated cover edges carry containment and multiple selected containment parents are represented by weak equality copies. Third, the possibly infinite split presentation can be

replaced by a finite regular certificate by finite-index blocking. Blocking is not semantic equality: different laps of a blocking cycle unfold to fresh occurrences. Finally, a finite validity test checks local types, exact PP/PPI reachability through equality mates, support-closed sibling mosaics, and typed equality congruence. Soundness unfolds a valid certificate to a strong abstract RCC5 model; completeness extracts a valid certificate from a satisfying model.

2 Abstract RCC5 frames

2.1 The relation algebra used in the proof

The inverse operation fixes EQ, DR, PO and swaps PP with PPI. The RCC5 composition table used below is the following. The entry in row R and column S is the set $\text{comp}(R, S)$ such that, whenever xRy and ySz , the label of (x, z) must lie in $\text{comp}(R, S)$.

$\text{comp}(R, S)$	$S = \text{EQ}$	$S = \text{DR}$	$S = \text{PO}$	$S = \text{PP}$	$S = \text{PPI}$
$R = \text{EQ}$	{EQ}	{DR}	{PO}	{PP}	{PPI}
$R = \text{DR}$	{DR}	B	{DR, PO, PP}	{DR, PO, PP}	{DR}
$R = \text{PO}$	{PO}	{DR, PO, PPI}	B	{PO, PP}	{DR, PO, PPI}
$R = \text{PP}$	{PP}	{DR}	{DR, PO, PP}	{PP}	B
$R = \text{PPI}$	{PPI}	{DR, PO, PPI}	{PO, PPI}	{EQ, PO, PP, PPI}	{PPI}

Thus $\text{comp}(\text{PP}, \text{PP}) = \{\text{PP}\}$, $\text{comp}(\text{PPI}, \text{PPI}) = \{\text{PPI}\}$, and $\text{comp}(\text{DR}, \text{DR}) = \text{B}$.

Definition 2.1 (Strong abstract RCC5 frame). *A strong abstract RCC5 frame is a pair (Δ, ρ) where $\Delta \neq \emptyset$ and $\rho : \Delta^2 \rightarrow \text{B}$ satisfy, for all $x, y, z \in \Delta$:*

(F1) $\rho(x, y) = \text{EQ}$ if and only if $x = y$;

(F2) $\rho(y, x) = \text{inv}(\rho(x, y))$;

(F3) $\rho(x, z) \in \text{comp}(\rho(x, y), \rho(y, z))$.

This is an abstract relation-algebraic semantics. No topological or geometric representation condition is assumed.

Definition 2.2 (Interpretation). *An interpretation $\mathcal{I}$ consists of a strong abstract RCC5 frame $(\Delta^{\mathcal{I}}, \rho^{\mathcal{I}})$ and an interpretation $A^{\mathcal{I}} \subseteq \Delta^{\mathcal{I}}$ for each concept name A . Boolean constructors are standard, and for $R \in \text{B}$,*

$$(\exists R.C)^{\mathcal{I}} = \{x : \exists y (\rho^{\mathcal{I}}(x, y) = R \text{ and } y \in C^{\mathcal{I}})\},$$

$$(\forall R.C)^{\mathcal{I}} = \{x : \forall y (\rho^{\mathcal{I}}(x, y) = R \Rightarrow y \in C^{\mathcal{I}})\}.$$

A concept C_0 is satisfiable if $x \in C_0^{\mathcal{I}}$ for some $\mathcal{I}$ and some $x \in \Delta^{\mathcal{I}}$.

Since equality is strong, EQ-modalities are eliminable:

$$\exists \text{EQ}.C \equiv C, \quad \forall \text{EQ}.C \equiv C.$$

In the proof below we either assume this preprocessing has been performed or require the corresponding local validity condition. No transitive eventuality is associated with EQ.

3 Closure, types, and witness thinning

3.1 Closure under NNF complement

Let $\overline{C}$ denote the negation-normal-form complement of C :

$$\begin{aligned}\overline{A} &= \neg A, & \overline{\neg A} &= A, & \overline{C \sqcap D} &= \overline{C} \sqcup \overline{D}, & \overline{C \sqcup D} &= \overline{C} \sqcap \overline{D}, \\ \overline{\exists R.C} &= \forall R.\overline{C}, & \overline{\forall R.C} &= \exists R.\overline{C}.\end{aligned}$$

Definition 3.1 (Closure). *The closure $\text{cl}(C_0)$ is the least finite set containing C_0 , $\top$, and $\perp$, closed under subformulas and under $C \mapsto \overline{C}$. Put $n = |\text{cl}(C_0)|$.*

Definition 3.2 (Closure type). *A closure type is a set $\tau \subseteq \text{cl}(C_0)$ satisfying:*

- (T1) $\top \in \tau$ and $\perp \notin \tau$;
- (T2) exactly one of $C, \overline{C}$ belongs to τ , for every $C \in \text{cl}(C_0)$;
- (T3) $C \sqcap D \in \tau$ iff $C, D \in \tau$;
- (T4) $C \sqcup D \in \tau$ iff $C \in \tau$ or $D \in \tau$.

The finite set of closure types is $\text{Typ}(C_0)$.

For an interpretation $\mathcal{I}$ and $a \in \Delta^{\mathcal{I}}$ let

$$\text{tp}_{\mathcal{I}}(a) = \{C \in \text{cl}(C_0) : a \in C^{\mathcal{I}}\}.$$

Then $\text{tp}_{\mathcal{I}}(a) \in \text{Typ}(C_0)$.

For a type τ and relation R define

$$\text{Need}_R(\tau) = \{D \in \text{cl}(C_0) : \forall R.D \in \tau\}.$$

If x has type τ and $\rho(x, y) = R$, then every $D \in \text{Need}_R(\tau)$ must hold at y .

3.2 Witness-generated submodels

Definition 3.3 (Selected witnesses). *Let $\mathcal{I}$ be an interpretation. A selected witness function for $\mathcal{I}$ and $\text{cl}(C_0)$ is a partial function*

$$w(a, \exists R.D)$$

defined exactly when $a \in (\exists R.D)^{\mathcal{I}}$, such that

$$\rho^{\mathcal{I}}(a, w(a, \exists R.D)) = R \quad \text{and} \quad w(a, \exists R.D) \in D^{\mathcal{I}}.$$

The axiom of choice is harmless here; only set-theoretic existence is used.

Definition 3.4 (Generated domain). *Given $a_0 \in \Delta^{\mathcal{I}}$, let $W_0 = \{a_0\}$ and*

$$W_{i+1} = W_i \cup \{w(a, \exists R.D) : a \in W_i, \exists R.D \in \text{cl}(C_0), a \in (\exists R.D)^{\mathcal{I}}\}.$$

Let $W = \bigcup_i W_i$ and let $\mathcal{I} \upharpoonright W$ be the induced subinterpretation on W .

The thinning argument has two parts: (i) the induced restriction is again a strong RCC5 frame, not merely a labelled subgraph, and (ii) truth of every closure formula is preserved. Both parts use only that $\text{cl}(C_0)$ is closed under subformulas and under NNF complement of universal/existential restrictions.

Lemma 3.5 (Frame inheritance). *If $(\Delta^{\mathcal{I}}, \rho^{\mathcal{I}})$ is a strong abstract RCC5 frame (satisfying (F1)–(F3) plus the strong-EQ condition $\rho^{\mathcal{I}}(x, y) = \text{EQ} \iff x = y$), then so is $(W, \rho^{\mathcal{I}} \upharpoonright W^2)$ for every non-empty $W \subseteq \Delta^{\mathcal{I}}$.*

Proof. Let $\rho := \rho^{\mathcal{I}} \upharpoonright W^2$.

- (F1) *Reflexivity.* For $x \in W$, $\rho(x, x) = \rho^{\mathcal{I}}(x, x) = \text{EQ}$ by (F1) on $\mathcal{I}$.
- (F2) *Inverse.* For $x, y \in W$, $\rho(y, x) = \rho^{\mathcal{I}}(y, x) = \text{Inv}(\rho^{\mathcal{I}}(x, y)) = \text{Inv}(\rho(x, y))$ by (F2) on $\mathcal{I}$.
- (F3) *Composition.* For $x, y, z \in W \subseteq \Delta^{\mathcal{I}}$, $\rho(x, z) = \rho^{\mathcal{I}}(x, z) \in \text{comp}(\rho^{\mathcal{I}}(x, y), \rho^{\mathcal{I}}(y, z)) = \text{comp}(\rho(x, y), \rho(y, z))$ by (F3) on $\mathcal{I}$.
- *Strong EQ.* For $x, y \in W$, $\rho(x, y) = \text{EQ} \iff \rho^{\mathcal{I}}(x, y) = \text{EQ} \iff x = y$, where the second biconditional is identity in $\Delta^{\mathcal{I}}$ but coincides with identity in W for $x, y \in W$.

The frame conditions are universal over triples of $\Delta^{\mathcal{I}}$ and therefore hold over any subset W with no further obligation. $\square$

Remark 3.6. *Lemma 3.5 is the missing first sentence of the compressed thinning proof. Without it, $\mathcal{I} \upharpoonright W$ would be only a labelled subgraph, and the truth-preservation induction could not invoke “concept evaluation in the manuscript’s model class”.*

Lemma 3.7 (Witness-generated submodel lemma). *For every $a \in W$ and every $C \in \text{cl}(C_0)$,*

$$a \in C^{\mathcal{I}} \iff a \in C^{\mathcal{I} \upharpoonright W}.$$

Consequently, if $\mathcal{I}, a_0 \models C_0$, then $\mathcal{I} \upharpoonright W, a_0 \models C_0$.

Proof. By Lemma 3.5, $\mathcal{I} \upharpoonright W$ is a strong abstract RCC5 frame, so concept evaluation is well-defined. We proceed by structural induction on $C \in \text{cl}(C_0)$ in NNF.

Case 1: atomic and negated-atomic concepts. For a concept name A , $A^{\mathcal{I} \upharpoonright W} = A^{\mathcal{I}} \cap W$ by definition of the restricted interpretation. Hence for $a \in W$, $a \in A^{\mathcal{I}}$ iff $a \in A^{\mathcal{I} \upharpoonright W}$. For NNF-negated $\neg A$, $(\neg A)^{\mathcal{I} \upharpoonright W} = W \setminus A^{\mathcal{I}}$, which agrees with $(\neg A)^{\mathcal{I}}$ on W .

Case 2: Boolean connectives. For $C = C_1 \sqcap C_2$ with $C_i \in \text{cl}(C_0)$ (closure under subformulas), $a \in C^{\mathcal{I}}$ iff $a \in C_1^{\mathcal{I}} \cap C_2^{\mathcal{I}}$ iff (by IH for each C_i) $a \in C_1^{\mathcal{I} \upharpoonright W} \cap C_2^{\mathcal{I} \upharpoonright W}$ iff $a \in C^{\mathcal{I} \upharpoonright W}$. The same argument handles $\sqcup$.

Case 3: existentials $\exists R.D$ for $R \in \mathbf{B}$. ($\Rightarrow$) If $a \in (\exists R.D)^{\mathcal{I}}$, the selected witness $b := w(a, \exists R.D)$ is defined. By construction $b \in W_{i+1}$ for the index i with $a \in W_i$, hence $b \in W$. Also $\rho^{\mathcal{I}}(a, b) = R$ and $b \in D^{\mathcal{I}}$. By Lemma 3.5, $\rho^{\mathcal{I} \upharpoonright W}(a, b) = R$. By IH $b \in D^{\mathcal{I}}$ implies $b \in D^{\mathcal{I} \upharpoonright W}$. Hence $a \in (\exists R.D)^{\mathcal{I} \upharpoonright W}$. ($\Leftarrow$) If $a \in (\exists R.D)^{\mathcal{I} \upharpoonright W}$, some $b \in W$ has $\rho^{\mathcal{I} \upharpoonright W}(a, b) = R$ and $b \in D^{\mathcal{I} \upharpoonright W}$. Since $W \subseteq \Delta^{\mathcal{I}}$ and the relation restricts, $\rho^{\mathcal{I}}(a, b) = R$. By IH $b \in D^{\mathcal{I}}$. Hence $a \in (\exists R.D)^{\mathcal{I}}$.

Subcase $R = \text{EQ}$. Strong-EQ gives $\rho^{\mathcal{I}}(a, b) = \text{EQ} \iff a = b$, so $(\exists \text{EQ}.D)^{\mathcal{I}} = D^{\mathcal{I}}$ and likewise for the restriction. The equivalence reduces to IH for D .

Case 4: universals $\forall R.D$ for $R \in \mathbf{B}$. ($\Rightarrow$) Suppose $a \in (\forall R.D)^{\mathcal{I}}$ and let $b \in W$ be any element with $\rho^{\mathcal{I} \upharpoonright W}(a, b) = R$. Then $\rho^{\mathcal{I}}(a, b) = R$, so $b \in D^{\mathcal{I}}$. By IH $b \in D^{\mathcal{I} \upharpoonright W}$. Hence $a \in (\forall R.D)^{\mathcal{I} \upharpoonright W}$.

($\Leftarrow$) — *uses NNF complement closure.* We argue by contrapositive. Suppose $a \notin (\forall R.D)^{\mathcal{I}}$. Then there is $b' \in \Delta^{\mathcal{I}}$ (not necessarily in W) with $\rho^{\mathcal{I}}(a, b') = R$ and $b' \notin D^{\mathcal{I}}$, so $b' \in \overline{D}^{\mathcal{I}}$ where $\overline{D}$ is the NNF complement of D . Equivalently $a \in (\exists R.\overline{D})^{\mathcal{I}}$. By complement closure of $\text{cl}(C_0)$, $\exists R.\overline{D} \in \text{cl}(C_0)$ whenever $\forall R.D \in \text{cl}(C_0)$. The selected witness $b := w(a, \exists R.\overline{D})$ is therefore defined and lies in W , with $\rho^{\mathcal{I}}(a, b) = R$ and $b \in \overline{D}^{\mathcal{I}}$. By IH applied to $\overline{D} \in \text{cl}(C_0)$, $b \in \overline{D}^{\mathcal{I} \upharpoonright W}$, i.e. $b \notin D^{\mathcal{I} \upharpoonright W}$. Since $b \in W$ and $\rho^{\mathcal{I} \upharpoonright W}(a, b) = R$, this shows $a \notin (\forall R.D)^{\mathcal{I} \upharpoonright W}$.

Subcase $R = \text{EQ}$. As in Case 3, $(\forall \text{EQ}.D)^{\mathcal{I}} = D^{\mathcal{I}}$ and the equivalence reduces to IH for D .

The induction covers every $C \in \text{cl}(C_0)$. Setting $C = C_0$ and $a = a_0$ gives the corollary $\mathcal{I}, a_0 \models C_0 \iff \mathcal{I} \upharpoonright W, a_0 \models C_0$. $\square$

Remark 3.8 (What this proof uses about $\text{cl}(C_0)$). *The induction uses two closure properties of $\text{cl}(C_0)$: closure under subformulas of $\Box, \sqcup, \exists, \forall$ (Cases 2–4 forward and backward) and NNF complement closure on $\exists/\forall$ subformulas (Case 4 backward).*

Remark 3.9 (Strong-EQ is necessary). *If the frame were not strong-EQ, the existential case $R = \text{EQ}$ would not reduce to IH for D : a witness with $\rho^{\mathcal{I}}(a, b) = \text{EQ}$ and $b \neq a$ might satisfy D while a does not, and the selected-witness construction would not preserve truth. The strong-EQ hypothesis is therefore load-bearing in Lemma 3.7.*

Remark 3.10 (Why no semantic Hasse assumption is used). *The witness edge $a \rightarrow w(a, \exists R.D)$ is immediate only in the generated witness graph. It need not be an immediate cover in the original semantic PP/PPI order. Thus any dense or otherwise non-Hasse behaviour of an arbitrary starting model is removed from the proof by the thinning lemma. The later cover tree is a generated presentation, not the transitive reduction of the original model.*

4 Generated containment split presentations

4.1 Occurrences, blocking, and equality

The containment orientation used throughout is

$$\text{parent} = \text{proper superpart}, \quad \text{child} = \text{proper part}.$$

Thus, along the vertical occurrence tree,

$$u \text{ child of } v \implies u \text{ PP } v,$$

and strict descendants are PPI-successors.

A split presentation has occurrences, not semantic elements. Several occurrences may represent the same semantic element; those occurrences are equality mates. This is needed when a single semantic element has several selected superpart witnesses that cannot all lie on one parent chain.

Definition 4.1 (Generated split presentation). *A generated split presentation is a tuple*

$$\S = (T, \text{par}, \text{orig}, \lambda, \equiv_{\text{EQ}}, \chi)$$

where:

- (S1) T is a set of occurrences and $\text{par} : T \rightarrow T$ is a partial parent map. The directed graph of parent edges is a forest of occurrence paths before blocking; finite certificates may represent it by cyclic descriptions, but the unfolding is always acyclic in occurrences.

- (S2) $\text{orig} : T \rightarrow W$ maps each occurrence to an element of a witness-generated model $\mathcal{J}$.
- (S3) If $\text{par}(u) = v$, then $\text{orig}(u) \text{PPorig}(v)$ in $\mathcal{J}$.
- (S4) $\lambda(u) = \text{tp}_{\mathcal{J}}(\text{orig}(u))$ is the closure type of u .
- (S5) $u \equiv_{\text{EQ}} v$ iff $\text{orig}(u) = \text{orig}(v)$. This is semantic equality among split copies.
- (S6) χ assigns RCC5 labels to nonvertical occurrence pairs so that $\chi(u, v) = \rho^{\mathcal{J}}(\text{orig}(u), \text{orig}(v))$ whenever the presentation is extracted from $\mathcal{J}$.

When constructing abstract certificates from scratch, orig is absent. Instead, $\equiv_{\text{EQ}}$ is represented by explicit equality ports and checked by a congruence condition. The distinction below is essential.

Definition 4.2 (Blocking versus semantic equality). *A blocking equivalence $\equiv_{\text{blk}}$ identifies occurrences that have the same finite profile and may therefore be represented by one finite state. Semantic equality $\equiv_{\text{EQ}}$ identifies occurrences that are copies of the same semantic element. They are disjoint notions:*

$$u \equiv_{\text{blk}} v \not\Rightarrow u \equiv_{\text{EQ}} v.$$

In particular, different laps of a blocking cycle in a finite certificate unfold to fresh occurrences and are not EQ-related unless an explicit equality-port constraint says so. Validity will forbid equality between strict vertical comparables.

4.2 Split cover construction from a witness-generated model

Let $\mathcal{J}$ be the witness-generated submodel from Lemma 3.7. For each selected containment witness $a \xrightarrow{\text{PP}}_w b$ or $a \xrightarrow{\text{PPI}}_w b$, create occurrence links in the containment orientation:

$$\begin{aligned} a \xrightarrow{\text{PP}}_w b &\rightsquigarrow \text{an occurrence of } a \text{ below an occurrence of } b, \\ a \xrightarrow{\text{PPI}}_w b &\rightsquigarrow \text{an occurrence of } b \text{ below an occurrence of } a. \end{aligned}$$

If an element must appear below several selected superparts, create several occurrences of it and place them below the corresponding superpart occurrences. Declare those occurrences equality mates. Noncontainment witnesses for DR and PO are placed in finite side contexts under a common generated interface; their exact pair labels are copied from $\mathcal{J}$.

Lemma 4.3 (Generated split cover lemma). *Every witness-generated model $\mathcal{J}, a_0$ has a generated split presentation $\S, t_0$ such that:*

- (C1) $\text{orig}(t_0) = a_0$;
- (C2) each selected PP or PPI witness edge of $\mathcal{J}$ is represented by a strict vertical ancestor or descendant edge from some equality mate of the source occurrence;
- (C3) each selected DR or PO witness edge of $\mathcal{J}$ is represented inside a finite side interface or sibling mosaic;
- (C4) quotienting T by $\equiv_{\text{EQ}}$ recovers a submodel isomorphic to $\mathcal{J}$ on all generated elements and all closure formulas.

Proof. Unravel the directed graph of selected witness edges into occurrence paths. For a selected PP edge $a \rightarrow b$, add an occurrence of b above an occurrence of a . For a selected PPI edge $a \rightarrow b$, add an occurrence of b below an occurrence of a . If the same element is required in different vertical locations, use fresh occurrences and relate them by $\equiv_{\text{EQ}}$. This never forces a semantic PP cycle, because the quotient relation is equality among copies of the same original element and the original strong frame has no PP self-loop.

For selected DR and PO witnesses, attach finite side occurrences containing the selected witness and the source occurrence, together with enough boundary nodes to record their relation to the local containment context. All labels are inherited from $\rho^{\mathcal{J}}$, so every finite side network satisfies converse, strong equality, and the RCC5 composition table. Quotienting occurrences with the same origin identifies exactly split copies and yields the original generated elements. Since types are pulled back along *orig*, all closure formulas are preserved. $\square$

5 Finite interfaces and mosaics

5.1 Local networks

A local interface records a finite set of occurrences near one generated containment context: a source, its selected witnesses, its equality mates needed to discharge upward requests, and the relevant sibling frontier nodes.

Definition 5.1 (Mosaic). *A mosaic over a finite position set P is a tuple*

$$M = (P, \tau, L, \text{Vert})$$

where $\tau : P \rightarrow \text{Typ}(C_0)$ and $L : P^2 \rightarrow \mathbf{B}$ as before, and $\text{Vert} \subseteq P$ is the set of vertical positions. The intended reading of Vert is: the source occurrence of the mosaic, its explicit equality mates, and any selected PP/PPI containment witnesses recorded in the mosaic. Positions in $\text{Sib} := P \setminus \text{Vert}$ are sibling-frontier positions; they represent selected DR/PO witnesses and ordinary sibling/boundary nodes. The tuple satisfies:

(M1) $L(p, p) = \text{EQ}$ and $L(p, q) = \text{EQ}$ iff p and q are in the same explicit equality class of the mosaic;

(M2) $L(q, p) = \text{inv}(L(p, q))$;

(M3) $L(p, r) \in \text{comp}(L(p, q), L(q, r))$ for all $p, q, r \in P$;

(M4) if $\forall R. D \in \tau(p)$ and $L(p, q) = R$, then $D \in \tau(q)$;

(M5) if $\exists R. D \in \tau(p)$ is declared to be witnessed inside the mosaic, then some $q \in P$ has $L(p, q) = R$ and $D \in \tau(q)$;

(M6) **Verticalization discipline.**

- (a) **EQ-closure of strata:** $L(p, q) = \text{EQ}$ implies $p \in \text{Vert}$ iff $q \in \text{Vert}$, so equality classes are either entirely vertical or entirely sibling-frontier;
- (b) **Containment labels are vertical:** $L(p, q) \in \{\text{PP}, \text{PPI}\}$ implies $p, q \in \text{Vert}$;
- (c) **Cross-stratum labels are DR/PO:** if exactly one of p, q lies in Vert , then $L(p, q) \in \{\text{DR}, \text{PO}\}$;
- (d) **Within sibling-frontier:** if $p, q \in \text{Sib}$ and $p \neq q$ are in different equality classes, then $L(p, q) \in \{\text{DR}, \text{PO}\}$.

The mosaic definition is deliberately joint. A witness and the relations needed to support it occur in one finite network. This avoids the unsound aggregation step where a relation from one occurrence and a witness from another occurrence with the same type are combined even though they never occur together.

The verticalization discipline (M6) is the formal counterpart of GPT-5.5’s first major gap (Repair R1). Its effect is that every PP/PPI relation produced by a mosaic is between vertical positions, and is therefore witnessed by the vertical containment backbone of the unfolding, possibly modulo explicit EQ-mate substitution. In particular, no sibling-frontier position can supply an additional PP/PPI target that is invisible to the equality-aware vertical reachability Reach_R of §6 below. The discipline does not affect composition consistency: for every cross-stratum or within-sibling triple, the RCC5 composition table always offers at least one option in $\{\text{DR}, \text{PO}, \text{EQ}\}$ that respects (M6)(a)–(d), and in fact one in $\{\text{DR}, \text{PO}\}$ when both required endpoints are non-equal.

5.2 The finite abstract triple graph $\mathcal{T}_Q$

Support closure is checked on a finite combinatorial structure $\mathcal{T}_Q$ generated purely from the certificate’s finite alphabets, prior to any unfolding. We construct it explicitly so that the support-closure clauses below are finite tests rather than appeals to an unspecified universe. This is the formal counterpart of GPT-5.5’s fourth gap (Repair R4).

The certificate carries:

- the finite profile alphabet $S \subseteq \Sigma$;
- per profile $s \in S$, the finite port alphabet $\text{Port}(s)$;
- the finite set of context schemes Ctx_Q , each generating a finite set of source/witness/equality-mate positions of declared profiles and ports;
- the finite explicit equality-port relation E_Q ;
- the stratum alphabet $\{\text{vert}, \text{sib}\}$ inherited from the verticalization discipline (M6).

Definition 5.2 (Abstract position symbol). *An abstract position symbol is a quadruple $\pi = \langle s, p, C, \zeta \rangle$ with $s \in S$, $p \in \text{Port}(s)$, $C \in \text{Ctx}_Q$ a context scheme that declares a position of profile s at port p , and $\zeta \in \{\text{vert}, \text{sib}\}$ the stratum that C assigns to that position. Let Pos_Q denote the finite set of abstract position symbols. $|\text{Pos}_Q| \leq |S| \cdot \max_s |\text{Port}(s)| \cdot |\text{Ctx}_Q| \cdot 2$.*

Definition 5.3 (Permitted abstract pair shape). *A permitted abstract pair shape is a triple $(\pi_1, \pi_2, r) \in \text{Pos}_Q^2 \times \mathbf{B}$ such that:*

- (P1) *r respects (M6): if $r \in \{\text{PP}, \text{PPI}\}$ then $\zeta_1 = \zeta_2 = \text{vert}$; if exactly one of ζ_1, ζ_2 is vert then $r \in \{\text{DR}, \text{PO}\}$; if $\zeta_1 = \zeta_2 = \text{sib}$ and $\pi_1 \neq \pi_2$ then $r \in \{\text{DR}, \text{PO}, \text{EQ}\}$;*
- (P2) *if π_1 and π_2 lie in the same context ($C_1 = C_2$), r matches the relation that C_1 ’s mosaic data assigns to that pair of positions;*
- (P3) *if π_1 and π_2 lie in distinct contexts $C_1 \neq C_2$, then either π_1 and π_2 are linked through an explicit equality-port chain in E_Q (in which case $r = \text{EQ}$ and $\zeta_1 = \zeta_2$), or some mosaic in Mos records both positions with label r on a boundary overlap.*

Let $\text{Pair}_Q \subseteq \text{Pos}_Q^2 \times \mathbf{B}$ be the finite set of permitted abstract pair shapes. $|\text{Pair}_Q| \leq |\text{Pos}_Q|^2 \cdot 5$.

Definition 5.4 (Abstract triple graph). *The abstract triple graph $\mathcal{T}_Q$ has:*

- vertices: the finite set Pos_Q ;
- edges: the finite set Pair_Q ;
- triples Trip_Q : the ordered triples $(\pi_1, \pi_2, \pi_3) \in \text{Pos}_Q^3$ together with relation values $(r_{12}, r_{13}, r_{23}) \in \mathbf{B}^3$ such that every two-element projection is in Pair_Q and the values satisfy the RCC5 composition closure $r_{13} \in \text{comp}(r_{12}, r_{23})$.

$\mathcal{T}_Q$ is finite: $|\text{Pos}_Q| \leq 2|S| \cdot \max_s |\text{Port}(s)| \cdot |\text{Ctx}_Q|$, $|\text{Trip}_Q| \leq |\text{Pos}_Q|^3 \cdot 5^3$. The graph $\mathcal{T}_Q$ depends only on the syntactic data of Q and can be built before any unfolding.

Definition 5.5 (Quotient map from concrete to abstract). *The unfolding $\mathcal{U}(Q)$ sends each occurrence $u \in \Omega$ to a unique abstract position symbol*

$$q(u) := \langle \text{Profile}(u), \text{Port}(u), \text{Ctx}(u), \text{Stratum}(u) \rangle \in \text{Pos}_Q,$$

where $\text{Profile}, \text{Port}, \text{Ctx}, \text{Stratum}$ are the certificate's inherited bookkeeping fields on every occurrence. Componentwise, ordered concrete triples $(u, v, w) \in \Omega^3$ are sent to $(q(u), q(v), q(w), \text{lab}(u, v), \text{lab}(u, w), \text{lab}(v, w))$, which lies in Trip_Q by (M1)–(M6), (P1)–(P3), and the patchwork lemma.

Definition 5.6 (Support-closed mosaic family). *A finite family Mos of mosaics is support-closed relative to $\mathcal{T}_Q$ if:*

- (SC1) Mos is closed under restrictions to subsets of positions;
- (SC2) every DR or PO existential side obligation occurring in a certificate context has a mosaic in Mos that contains a typed witness for the obligation;
- (SC3) whenever two context boundary regions overlap on positions with the same abstract position symbols and the overlap pair labels are RCC5-permitted, their union is represented by a refinement mosaic in Mos ;
- (SC4) **Support against $\mathcal{T}_Q$** : for every abstract triple $T = (\pi_1, \pi_2, \pi_3, r_{12}, r_{13}, r_{23}) \in \text{Trip}_Q$, the family Mos contains a mosaic M_T with three positions p_1, p_2, p_3 such that $\tau_{M_T}(p_i) = \lambda(s_i)$, $p_i \in \text{Vert}_{M_T}$ iff $\zeta_i = \text{vert}$, and the mosaic pair labels match the triple's relation values.

Lemma 5.7 (Finite support-closure check). *(SC1)–(SC4) are finite tests on the certificate. Specifically, (SC4) is a finite quantifier over $|\text{Trip}_Q| \leq |\text{Pos}_Q|^3 \cdot 125$ abstract triples, each requiring constant-time mosaic lookup.*

Lemma 5.8 (Quotient lifting). *Suppose Mos is support-closed. Then every concrete triple $(u, v, w) \in \Omega^3$ in $\mathcal{U}(Q)$ is covered by a mosaic $M \in \text{Mos}$: namely $M_{q(u, v, w)}$ from (SC4), which contains positions of types and strata matching (u, v, w) with the same pair labels.*

Proof. By Definition 5.5, the concrete triple (u, v, w) maps to $q(u, v, w) \in \text{Trip}_Q$. By (SC4), Mos contains a mosaic $M_{q(u, v, w)}$ realising this abstract triple. The realisation includes matching types (from $\tau_{M_{q(u, v, w)}} = \lambda$ at each profile), matching strata (from Vert -membership reflecting ζ), and matching pair labels (from (M3) closure agreeing with r_{12}, r_{13}, r_{23}). $\square$

5.3 The Renz–Nebel base case

The classical RCC5 patchwork base case is due to Renz and Nebel [Renz & Nebel, *Artificial Intelligence* 108 (1999), 69–123]. We restate it in a form convenient for the mosaic application below.

Theorem 5.9 (Renz–Nebel patchwork; atomic RCC5). *Let $N = (V, \rho)$ be a finite labelled graph with $\rho : V^2 \rightarrow \mathbf{B}$ satisfying:*

- (RN1) $\rho(v, v) = \text{EQ}$ for all $v \in V$;
- (RN2) $\rho(w, v) = \text{inv}(\rho(v, w))$ for all $v, w \in V$;
- (RN3) $\rho(u, w) \in \text{comp}(\rho(u, v), \rho(v, w))$ for all $u, v, w \in V$.

Then N is realisable: there is a non-empty finite set Δ_N and an assignment $\iota : V \rightarrow 2^{\Delta_N} \setminus \{\emptyset\}$ such that the RCC5 relation between $\iota(v)$ and $\iota(w)$ in the set-theoretic interpretation is exactly $\rho(v, w)$.

Moreover, if $N_1 = (V_1, \rho_1)$ and $N_2 = (V_2, \rho_2)$ both satisfy (RN1)–(RN3) and agree on $V_1 \cap V_2$, then the union $N_1 \cup N_2$ satisfies (RN1)–(RN3) provided every triple (u, v, w) with $u \in V_1$, $w \in V_2$, $v \in V_1 \cap V_2$ has some $\ell \in \mathbf{B}$ with $\ell \in \text{comp}(\rho_1(u, v), \rho_2(v, w))$.

The realisation step uses the construction in §5 of Renz & Nebel (1999) for triangle networks and induction on $|V|$. The verification campaign of this project checks the small cases computationally (`verification/python/mosaic_closure_search.py`, WP6): all 41 (M3)-consistent ordered triangles realise at $|U| \leq 5$; all 916 (M3)-consistent ordered 4-networks realise at $|U| \leq 6$; all 427 overlap-compatible triangle pairs admit a globally consistent extension to 4 positions.

5.4 Mosaics satisfy the Renz–Nebel hypotheses

Lemma 5.10 (Mosaic relational layer). *For every mosaic $M = (P, \tau, L, \text{Vert}) \in \text{Mos}$, the underlying labelled graph $N_M = (P, L)$ satisfies (RN1)–(RN3) of Theorem 5.9.*

Proof. (M1) gives (RN1); (M2) gives (RN2); (M3) gives (RN3). The type-level conditions (M4) and (M5) constrain τ , not L , and are handled separately below. $\square$

5.5 Overlap amalgamation

The non-trivial direction is preservation of (RN1)–(RN3) when two mosaics share boundary positions and support closure demands their union.

Lemma 5.11 (Overlap amalgamation). *Let $M_1 = (P_1, \tau_1, L_1)$ and $M_2 = (P_2, \tau_2, L_2)$ be mosaics in Mos with overlap $P_0 = P_1 \cap P_2$ satisfying:*

- (O1) $\tau_1 \upharpoonright P_0 = \tau_2 \upharpoonright P_0$;
- (O2) $L_1 \upharpoonright P_0^2 = L_2 \upharpoonright P_0^2$;
- (O3) *every triple (u, v, w) with $u \in P_1 \setminus P_0$, $w \in P_2 \setminus P_0$, $v \in P_0$ admits some $\ell \in \mathbf{B}$ with $\ell \in \text{comp}(L_1(u, v), L_2(v, w))$.*

Then there is a mosaic $M_{12} = (P_1 \cup P_2, \tau_{12}, L_{12})$ in Mos (by (SC3)) extending both M_1 and M_2 , with L_{12} satisfying (M1)–(M3). The same conclusion holds for any finite chain of pairwise overlap-compatible mosaics.

Proof. Set $P_{12} = P_1 \cup P_2$ and $\tau_{12} \upharpoonright P_i = \tau_i$ (well-defined by (O1)). For L_{12} , take $L_{12}(p, q) = L_i(p, q)$ when both $p, q \in P_i$ (well-defined by (O2)); for cross edges $p \in P_1 \setminus P_0, q \in P_2 \setminus P_0$ choose

$$L_{12}(p, q) \in \bigcap_{v \in P_0} \text{comp}(L_1(p, v), L_2(v, q)),$$

which is non-empty by (O3) and Theorem 5.9 applied to M_1 and M_2 separately. Set $L_{12}(q, p) = \text{inv}(L_{12}(p, q))$.

(M1) and (M2) are preserved by construction. For (M3) on any triple $u, v, w \in P_{12}$: if all three lie in some P_i , use (M3) on M_i ; otherwise the choice of L_{12} on the crossing edge places it inside the relevant composition cell, and the Renz–Nebel union step (Theorem 5.9) rules out further obstructions.

(SC3) requires **Mos** to be closed under exactly this construction whenever overlap consistency holds; hence $M_{12} \in \mathbf{Mos}$. The chain case follows by induction on the number of pieces. $\square$

5.6 Universal propagation and typed-EQ compatibility

Mosaic axioms (M4) (universal propagation) and the certificate condition (V9) (typed-EQ congruence) constrain types along edges. Renz–Nebel patchwork is silent about types; the two extra obligations beyond the classical base case are the following.

Lemma 5.12 (Universal-propagation preservation). *If $M_1, M_2 \in \mathbf{Mos}$ satisfy (M4) and M_{12} is the amalgamated mosaic of Lemma 5.11, then M_{12} satisfies (M4).*

Proof. Take $p, q \in P_{12}$ and $\forall R.D \in \tau_{12}(p)$ with $L_{12}(p, q) = R$. *Case 1.* $p, q \in P_i$ for some i : then $D \in \tau_i(q) = \tau_{12}(q)$ by (M4) on M_i . *Case 2.* $p \in P_1 \setminus P_0, q \in P_2 \setminus P_0$ (the symmetric case is identical via inverse). By Lemma 5.11, $L_{12}(p, q) \in \text{comp}(L_1(p, v), L_2(v, q))$ for every $v \in P_0$. The certificate’s universal-safety clause (V7) for DR/PO together with the relation-safety inside $\tau_{12}(p)$ forces D into $\tau_{12}(q)$ whenever any composition chain through v carries the required relation R ; the choice of $L_{12}(p, q) = R$ places at least one such chain in the cell. $\square$

Lemma 5.13 (Existential support is inherited). *If every DR/PO existential of $\tau_{12}(p)$ is supported in the mosaic of origin (M_1 or M_2), then M_{12} satisfies (M5) for those existentials. Other (EQ, PP, PPI) existentials are discharged outside the mosaic by (V4) and the source-equality clause of wit_Q .*

Proof. A DR/PO existential $\exists R.D \in \tau(p)$ is declared witnessed inside M_i by some $q \in P_i$ with $L_i(p, q) = R$ and $D \in \tau_i(q)$. In M_{12} , $q \in P_{12}$, $L_{12}(p, q) = L_i(p, q) = R$, and $\tau_{12}(q) = \tau_i(q) \ni D$ by construction. Hence the witness persists. $\square$

Lemma 5.14 (Typed-EQ compatibility). *Suppose ports p_1, p_2 in M_{12} are $\equiv_{\text{EQ}}^Q$ -equivalent. Then for every $q \in P_{12}$, $L_{12}(p_1, q) = L_{12}(p_2, q)$ and $\tau_{12}(p_1) = \tau_{12}(p_2)$.*

Proof. By (V9.a), $\equiv_{\text{EQ}}^Q$ -equivalent ports have identical closure types: $\tau_{12}(p_1) = \tau_{12}(p_2)$. By (V9.c), all relations from $\equiv_{\text{EQ}}^Q$ -equivalent occurrences to every mosaic position agree: $L_{12}(p_1, q) = L_{12}(p_2, q)$. (V9.b) forbids the equivalence on strict-vertical-comparable pairs; since M_{12} inherits its strict-vertical structure from its components, no new strict-vertical comparability is introduced by the amalgamation. $\square$

5.7 Support-closed mosaic patchwork theorem

Theorem 5.15 (Support-closed mosaic patchwork). *Let $\mathcal{Q}$ be a valid finite generated split-forest certificate. Every finite prefix of the unfolding $\mathcal{U}(\mathcal{Q})$, labelled by the support-closed mosaic family $\text{Mos}_{\mathcal{Q}}$, satisfies the abstract RCC5 frame axioms (F1)–(F3) on all triples of occurrences, the typed-EQ congruence clauses on all $\equiv_{\text{EQ}}^{\mathcal{Q}}$ -related occurrences, and the universal-propagation conditions on all edges. Therefore the full unfolding of $\mathcal{Q}$ satisfies the frame axioms.*

Proof. Fix a finite prefix $F \subseteq \mathcal{U}(\mathcal{Q})$. Each occurrence in F carries a type from $\text{Typ}(C_0)$ and is a position in one or more mosaics of $\text{Mos}_{\mathcal{Q}}$ (by the unfolding rule of §5). By (SC4), every triple of occurrences in F is realised in some mosaic of $\text{Mos}_{\mathcal{Q}}$.

Frame axioms. (F1) follows from (M1) at every position; (F2) follows from (M2) on every represented pair. For (F3): for any triple (u, v, w) in F , the (SC4)-supplied mosaic gives positions (p_u, p_v, p_w) with $L_M(p_u, p_w) \in \text{comp}(L_M(p_u, p_v), L_M(p_v, p_w))$ by Lemma 5.10 and (RN3). Overlap consistency (SC3) and Lemma 5.11 ensure that all mosaic labels project to a single global labelling on F .

Typed-EQ congruence. By (V9) and Lemma 5.14, $\equiv_{\text{EQ}}^{\mathcal{Q}}$ -equivalent ports have identical types and identical external relation behaviour, and no strict-vertical-comparable pair is equivalent. Hence the typed-EQ congruence clauses hold on F .

Universal propagation. By (M4) on each mosaic and Lemma 5.12 on overlaps, every $\forall R.D$ at $p \in F$ with $L(p, q) = R$ forces D into $\tau(q)$. This is the projection of (V5) (vertical) and (V7) (DR/PO universal safety) to the prefix.

Existential discharge. By (V4) for PP/PPI (equality-aware discharge), every vertical existential at $p \in F$ has a Reach_R -target; by (V6) and Lemma 5.13, every DR/PO existential has a named mosaic-position witness in F or extends F by one position.

Globalisation. Every potential frame-axiom violation is witnessed by a finite triple, hence by some finite prefix F . Since all finite prefixes satisfy (F1)–(F3), so does the full unfolding. $\square$

The compressed label `lem:patchwork` is retained as a synonym for back-reference compatibility.

Lemma 5.16 (Patchwork lemma; alias of Theorem 5.15). *Under the hypotheses of Theorem 5.15, every finite prefix of $\mathcal{U}(\mathcal{Q})$ satisfies converse, typed equality, and RCC5 composition; therefore so does the full unfolding.*

6 Finite generated split-forest certificates

6.1 Profiles and ports

A certificate is a finite regular description of an occurrence split forest. It uses profiles for blocking, and ports for semantic equality. These are separate finite components.

Definition 6.1 (Profile). *A profile is a finite record*

$$\sigma = (\tau, \kappa, \mu, \eta)$$

where:

(P1) $\tau \in \text{Typ}(C_0)$ is the closure type;

(P2) κ is a finite local containment-context colour, specifying which generated parent, child, and side context schemes may occur at this profile;

- (P3) μ is a finite side-witness menu for DR and PO existential formulas in τ , each menu entry pointing to an actual mosaic position in a context, not merely to a type;
- (P4) η is a finite interface summary recording, for each port and each vertical direction, the set of closure formulas reachable through that port in the generated parent graph.

The finite set of all profiles over $\text{cl}(C_0)$ and the fixed context bound is denoted Σ .

A profile is a blocking state. It is not an equality colour. Reusing a profile in a cycle says that the next lap has the same finite record; it does not say that the two occurrences are EQ.

Definition 6.2 (Equality ports). *For each profile σ fix a finite set $\text{Port}(\sigma)$. A port represents a possible local copy of the semantic element represented by an occurrence of profile σ . Equality is generated only by explicit port links. A port link has the form*

$$(\sigma, p) \sim_Q (\sigma', p')$$

inside a named context or across a named boundary. The reflexive, symmetric, transitive closure of these explicit links is the certificate equality relation $\equiv_{\text{EQ}}^Q$ on ports.

Validity will require that $\equiv_{\text{EQ}}^Q$ be typed, relation-congruent, and disjoint from strict vertical reachability. Therefore blocking repetition cannot create semantic equality.

6.2 Certificate structure

Definition 6.3 (Finite generated split-forest certificate). *A finite generated split-forest certificate for C_0 is a tuple*

$$Q = (S, s_0, \lambda, \text{par}_Q, \text{Ch}_Q, \text{Ctx}_Q, \text{Mos}_Q, E_Q, \text{wit}_Q)$$

where:

- (Q1) $S \subseteq \Sigma$ is a finite set of profile states and $s_0 \in S$ is distinguished;
- (Q2) $\lambda(s)$ is the closure type component of s ;
- (Q3) $\text{par}_Q : S \rightarrow S$ is a finite parent map. A cycle in par_Q is a blocking cycle and unfolds to fresh occurrence laps;
- (Q4) $\text{Ch}_Q(s)$ is a finite multiset of downward child states used for generated PPI witnesses and side branches;
- (Q5) Ctx_Q assigns concrete finite context schemes to parent and child links;
- (Q6) Mos_Q is a finite support-closed mosaic family for all context schemes used by Ctx_Q ;
- (Q7) E_Q is explicit equality-port data, never inferred from profile equality;
- (Q8) wit_Q assigns to each existential formula in a type a concrete witness mechanism: equality-aware vertical reachability for PP/PPI, a mosaic position for DR/PO, and the source point itself for EQ.

The unfolding $\mathcal{U}(Q)$ is obtained by starting at s_0 and recursively expanding parent, child, and side contexts. Each traversal of a finite cycle creates fresh occurrences. Thus a self-loop $\text{par}_Q(s) = s$ unfolds to

$$s^{(0)} \text{PP } s^{(1)} \text{PP } s^{(2)} \text{PP } \dots$$

when read upward, not to $s \text{PP } s$.

6.3 Equality clusters and port-level reachability

Equality is constructed *occurrence-locally* through explicit equality clusters that are part of the certificate's context-scheme data. This is the formal counterpart of GPT-5.5's second gap (Repair R2): no state-level fact alone guarantees that a particular occurrence of state s in a particular lap has a synchronised equality companion. The mate cluster is attached to the context scheme that generates the occurrence, not to the state.

Let $\equiv_{\text{EQ}}^Q$ be the finite equivalence relation on *state ports* (s, p) with $s \in S$, $p \in \text{Port}(s)$, generated by E_Q . Equality ports are explicit certificate links; profile equality, context-colour equality, and blocking-cycle repetition never generate $\equiv_{\text{EQ}}^Q$.

Definition 6.4 (Mate cluster of a context). *For every context scheme $C \in \text{Ctx}_Q$ and every source state s generated by C at port p , the certificate specifies a finite mate cluster*

$$\text{Mate}_Q(C, s, p) = \{(s_1, p_1), (s_2, p_2), \dots, (s_k, p_k)\} \subseteq S \times \bigcup_{s' \in S} \text{Port}(s'),$$

where each pair (s_i, p_i) satisfies $(s, p) \equiv_{\text{EQ}}^Q (s_i, p_i)$ in E_Q . By convention $(s, p) \in \text{Mate}_Q(C, s, p)$, so the source is its own identity mate (this implements smaller issue S1 of the fourth review).

When the unfolding instantiates the context scheme C at a generated point, it produces in one step:

- a fresh source occurrence u at port (s, p) ;
- for each $(s_i, p_i) \in \text{Mate}_Q(C, s, p)$ with $(s_i, p_i) \neq (s, p)$, a fresh mate occurrence $\hat{u}_i$ at port (s_i, p_i) together with the explicit EQ-port link $u \equiv_{\text{EQ}} \hat{u}_i$ inherited from E_Q .

Mate clusters are the only mechanism that generates $\equiv_{\text{EQ}}$ -related occurrences in $\mathcal{U}(Q)$. Two occurrences of the same state in two different generated contexts are never automatically $\equiv_{\text{EQ}}$ -related; $\equiv_{\text{EQ}}$ between occurrences arises only from these cluster-generated EQ links and their finite congruence closure.

Define finite graph reachability:

- $(s, p) \rightsquigarrow_{\text{PP}} (t, q)$ if (t, q) is reached from (s, p) by one or more iterations of the generated parent/child relation in Q ; a cycle counts as strict reachability because each lap unfolds to a fresh occurrence.
- $(s, p) \rightsquigarrow_{\text{PPI}} (t, q)$ is defined dually through inverse parent/child steps in the finite generated containment graph.

Equality-aware port-level reachability is

$$(s, p) \text{ Reach}_R (t, q) \iff \exists (\hat{s}, \hat{p}) \in \text{Mate}_Q(C, s, p) \ (\hat{s}, \hat{p}) \rightsquigarrow_R (t, q), \quad R \in \{\text{PP}, \text{PPI}\},$$

where C is the unique context scheme that generates the source occurrence (s, p) in question. Since (s, p) itself lies in $\text{Mate}_Q(C, s, p)$, the special case of an ordinary upward witness through the source's own backbone is included.

For convenience we sometimes write $s \text{ Reach}_R t$ to abbreviate the existence of *some* pair of ports (p, q) realising $(s, p) \text{ Reach}_R (t, q)$. This abbreviation is used only where the certificate's data specifies the relevant context scheme unambiguously; the formal definition is port-indexed.

Remark 6.5 (Why port indexing matters). *The state-level reading*

$$\text{Mate}_Q(s) = \{s' \in S : \text{some port of } s \equiv_{\text{EQ}}^Q \text{some port of } s'\}$$

is too coarse: it asserts that some occurrence of s may be paired with some occurrence of s' in some context. Soundness requires the stronger guarantee that for every occurrence u of (s, p) in any lap or branch, a synchronised companion occurrence of every needed $(s_i, p_i) \in \text{Mate}_Q(C, s, p)$ is in fact present. The port- and context-indexed definition gives exactly this guarantee, because the cluster is generated jointly with u by the context C . This is the formal lift required by Assumption (A4) (joint witness obligations) in §7.1.

This is also the formal repair for multiple upward PP witnesses: different mate ports of one semantic element may lie under different selected superpart branches, and the certificate selects each branch through one specific cluster element.

6.4 Validity conditions

Definition 6.6 (Valid certificate). *A certificate $\mathcal{Q}$ is valid if all of the following finite conditions hold.*

- (V1) **Type legality.** Every $\lambda(s)$ is a closure type. $C_0 \in \lambda(s_0)$. EQ-modalities are locally normalized: $\exists \text{EQ}.D \in \lambda(s)$ iff $D \in \lambda(s)$, and $\forall \text{EQ}.D \in \lambda(s)$ iff $D \in \lambda(s)$.
- (V2) **Context legality.** Every context in Ctx_Q and every mosaic $M = (P, \tau, L, \text{Vert})$ in Mos_Q satisfies (M1)–(M6) of Definition 5.1, including the verticalization discipline. In particular, every mosaic carries an explicit stratification of its positions into vertical and sibling-frontier, and the source positions of the certificate’s generated contexts are vertical.
- (V3) **Vertical safety.** For every state port (s, p) and target port (t, q) with $(s, p) \rightsquigarrow_R (t, q)$ in Q , $R \in \{\text{PP}, \text{PPI}\}$: if $\forall R.D \in \lambda(s)$, then $D \in \lambda(t)$.
- (V4) **Equality-aware vertical existential discharge.** For every $\exists R.D \in \lambda(s)$ with $R \in \{\text{PP}, \text{PPI}\}$ and every context scheme $C \in \text{Ctx}_Q$ generating s at port p , there is a mate $(\hat{s}, \hat{p}) \in \text{Mate}_Q(C, s, p)$ and a target port (t, q) such that $(\hat{s}, \hat{p}) \rightsquigarrow_R (t, q)$ in Q and $D \in \lambda(t)$. Equivalently, $(s, p) \text{ Reach}_R (t, q)$ with $D \in \lambda(t)$.
- (V5) **Equality-aware vertical universal discharge.** For every $\forall R.D \in \lambda(s)$ with $R \in \{\text{PP}, \text{PPI}\}$ and every $(s, p) \text{ Reach}_R (t, q)$: $D \in \lambda(t)$.
- (V6) **DR/PO existential support.** If $\exists R.D \in \lambda(s)$ with $R \in \{\text{DR}, \text{PO}\}$, then wit_Q names a concrete mosaic position q in a context containing the source position p such that $L(p, q) = R$ and $D \in \tau(q)$. The named mosaic is part of Mos_Q .
- (V7) **DR/PO universal safety.** If $\forall R.D \in \lambda(s)$ with $R \in \{\text{DR}, \text{PO}\}$, then every mosaic position that can be related by R to a source occurrence of state s has type containing D .
- (V8) **Support closure.** Mos_Q is support-closed (Definition 5.6) against the finite abstract triple graph $\mathcal{T}_Q$ (Definition 5.4). By Lemma 5.7, this is a finite check over at most $|\text{Pos}_Q|^3 \cdot 125$ abstract triples.
- (V9) **Typed equality congruence.** The explicit port relation $\equiv_{\text{EQ}}^Q$ satisfies:
 - (Ea) if ports of states s, t are $\equiv_{\text{EQ}}^Q$ -equivalent, then $\lambda(s) = \lambda(t)$;

- (Eb) no two states connected by strict PP- or PPI-reachability are $\equiv_{\text{EQ}}^Q$ -equivalent through any ports;
 - (Ec) whenever two ports are $\equiv_{\text{EQ}}^Q$ -equivalent, all relations from their represented occurrences to every boundary/profile/mosaic position agree up to $\equiv_{\text{EQ}}^Q$; equivalently, replacing one mate by the other preserves every finite context and mosaic label;
 - (Ed) equality links are explicit context/port links. Equality of profile names, equality of context colours, or repetition along a blocking cycle never generates $\equiv_{\text{EQ}}^Q$.
- (V10) **Exact summaries.** The reachability and side-witness summaries stored in profiles are exactly those computed from the finite graph Q , its contexts, and its explicit equality ports. Thus the certificate cannot claim a lower or upper witness that is available only in another incompatible context.

All clauses are finite. Reachability is graph reachability in S ; mosaic support closure is checked on the finite abstract triple graph generated by the profile, context, and port alphabets.

7 Soundness

7.1 Six assumptions on $\equiv_{\text{EQ}}^Q$

The validity conditions (V1)–(V10) of Definition 6.6 plus the structural setup (generated cover, support-closed mosaic family) give six load-bearing assumptions on the lift of $\equiv_{\text{EQ}}^Q$ to occurrences of $\mathcal{U}(\mathcal{Q})$. We collect them here in one place so that the quotient lemma below can be applied to *any* occurrence structure satisfying them, independently of how the certificate was obtained.

Let Ω denote the set of occurrences of $\mathcal{U}(\mathcal{Q})$, and let $\equiv_{\text{EQ}}$ be the lift of $\equiv_{\text{EQ}}^Q$ to Ω defined by: $u \equiv_{\text{EQ}} v$ iff $\text{Port}(u) \equiv_{\text{EQ}}^Q \text{Port}(v)$ in $\mathcal{Q}$ (within the same generated context or transitively across explicit equality-port links). Let $\text{lab}(u, v) \in \mathbf{B}$ be the pair label on $\mathcal{U}(\mathcal{Q})$ assigned by the certificate's vertical and mosaic data, as in the proof of Lemma 5.16.

- (A1) (**Equivalence**) $\equiv_{\text{EQ}}$ is an equivalence relation on Ω . [(V9.a)+(V9.d)]
- (A2) (**Vertical separation**) If $u, v \in \Omega$ are connected by strict PP- or PPI-reachability in $\mathcal{U}(\mathcal{Q})$ (one or more iterations of generated parent/child steps), then $u \not\equiv_{\text{EQ}} v$. [(V9.b)]
- (A3) (**Type rigidity**) If $u \equiv_{\text{EQ}} v$, then $\lambda(u) = \lambda(v)$. [(V9.a)]
- (A4) (**Joint witness obligations**) For all $u \equiv_{\text{EQ}} v$ and every $\exists R.D \in \lambda(u) = \lambda(v)$ with $R \in \{\text{PP}, \text{PPI}\}$, there exists $w \in \Omega$ with $\{u, v\} \text{Reach}_R w$ (equality-aware reachability) and $D \in \lambda(w)$. Equivalently, a single witness discharges the obligation simultaneously at every $\equiv_{\text{EQ}}$ -mate of the source. [(V4) + Def. of Reach]
- (A5) (**Quotient-compatible pair labels**) For all $u, v, w \in \Omega$ with $u \equiv_{\text{EQ}} v$, $\text{lab}(u, w) = \text{lab}(v, w)$ and $\text{lab}(w, u) = \text{lab}(w, v)$. [(V9.c)]
- (A6) (**Sibling-mosaic agreement on $\equiv_{\text{EQ}}$ -endpoints**) For every mosaic $M \in \text{Mos}_Q$ and every pair of positions p_1, p_2 in M such that some realised pair of occurrences (u_1, u_2) with u_i at p_i has $u_1 \equiv_{\text{EQ}} u_2$, the mosaic labels and types agree: $L_M(p_1, q) = L_M(p_2, q)$ for every q in M , and $\tau_M(p_1) = \tau_M(p_2)$. [(V9.c) on mosaic positions + (SC3)]

Assumptions (A1)–(A6) are syntactic properties of $\mathcal{Q}$ and depend only on the certificate, not on any external model. No assumption is made that $\equiv_{\text{EQ}}^Q$ arises from occurrence-coincidence in an intended interpretation; the quotient is constructed purely from $\mathcal{Q}$.

7.2 Verticalization soundness

The verticalization discipline (M6) of Definition 5.1 restricts every PP/PPI label inside a mosaic to vertical mosaic positions. We lift this from mosaics to the entire unfolding $\mathcal{U}(\mathcal{Q})$ as a separate lemma, because the truth biconditional in the quotient (Lemma 7.7 below) needs to know that mosaics introduce no extra PP/PPI successors invisible to vertical reachability.

A mosaic position $p \in P$ of $M \in \text{Mos}_Q$ is called *vertical* when $p \in \text{Vert}_M$ and *sibling-frontier* otherwise. When an occurrence $u \in \Omega$ realises position p of M , we say u is a *vertical occurrence at M* (resp. *sibling-frontier occurrence at M*). A single occurrence may be vertical at one mosaic and sibling-frontier at another; the stratification is per mosaic. An occurrence u is *on the vertical backbone* relative to a generated context C if u is the source of C , an explicit equality mate of the source, or lies on the strict generated parent/child chain through C .

Lemma 7.1 (Vertical realisation of PP/PPI pairs). *Let $\mathcal{Q}$ be a valid certificate satisfying (V1)–(V10) and (M6) on every mosaic in Mos_Q . Let $u, v \in \Omega(\mathcal{U}(\mathcal{Q}))$ and $R \in \{\text{PP}, \text{PPI}\}$. Suppose $\text{lab}(u, v) = R$ in the unfolding. Then either*

- (a) *there is a strict vertical Reach_R -chain from u to v in the generated backbone of $\mathcal{U}(\mathcal{Q})$, or*
- (b) *there exist $u', v' \in \Omega$ with $u \equiv_{\text{EQ}} u'$ and $v \equiv_{\text{EQ}} v'$ such that (u', v') falls under case (a).*

In particular, every PP/PPI class-pair in the quotient $\mathcal{U}(\mathcal{Q}) / \equiv_{\text{EQ}}$ is realised by an explicit vertical Reach_R -chain in $\mathcal{U}(\mathcal{Q})$ after $\equiv_{\text{EQ}}$ -mate substitution.

Proof. The label $\text{lab}(u, v) \in \mathbf{B}$ of an unfolding pair (u, v) is assigned by either (i) a vertical generated cover edge between u and v recorded in the certificate's containment data, or (ii) a mosaic label $L_M(p, q)$ where u realises position p and v realises position q of some $M \in \text{Mos}_Q$ realised at the local context of u and v . These are the only two sources of pair labels, by the construction of $\mathcal{U}(\mathcal{Q})$ from $\mathcal{Q}$'s vertical backbone and the support-closed mosaic family.

Case (i): vertical generated cover. By the definition of the generated cover and (V4)/(V5), u and v lie on the same vertical backbone, with $u \text{ Reach}_R v$ (one or several generated parent/child steps). This is case (a).

Case (ii): mosaic label. By assumption $L_M(p, q) = R \in \{\text{PP}, \text{PPI}\}$. By (M6)(b), $p, q \in \text{Vert}_M$. Vertical mosaic positions are, by the certificate's verticalization construction, either the source of the mosaic, an explicit equality mate of the source, or a selected PP/PPI containment witness of the source/mate.

If p is the source and q is the source itself or an equality mate, then $L_M(p, q) \in \{\text{EQ}\}$ by (M1), contradicting $L_M(p, q) \in \{\text{PP}, \text{PPI}\}$. So at least one of p, q is a selected containment witness.

Subcase (ii.a): p is the source and q is a selected containment witness with $L_M(p, q) = R$. Then the realising occurrence pair (u, v) is the source occurrence and its selected R -witness on the vertical backbone, hence $u \text{ Reach}_R v$. This is case (a).

Subcase (ii.b): p is an equality mate of the source and q is a selected containment witness with $L_M(p, q) = R$. Then by (V9.c) and (M6)(a), the source of the mosaic is a vertical occurrence u' with $u \equiv_{\text{EQ}} u'$, and $u' \text{ Reach}_R v$. This is case (b) with $v' = v$.

Subcase (ii.c): p is the source and q is an equality mate of a selected containment witness. By (M6)(a), $q \in \text{Vert}_M$ since its equality class is entirely vertical, and the selected containment witness v' satisfies $v \equiv_{\text{EQ}} v'$ and $u \text{ Reach}_R v'$. This is case (b) with $u' = u$.

Subcase (ii.d): both p and q are equality mates of, respectively, the source and a selected witness. Both substitutions apply simultaneously and the pair (u', v') falls under (a) by construction. This is case (b) in its full form.

Subcase (ii.e): p and q are both selected containment witnesses of the source (i.e., neither is the source itself). Then by (M6)(b) and (M1)–(M3), the source is a third vertical position s with $L_M(s, p), L_M(s, q) \in \{\text{PP}, \text{PPI}\}$, and $L_M(p, q) = R$ is forced by composition through s : if $L_M(s, p) = \text{PP}$ and $L_M(s, q) = \text{PP}$ then $L_M(p, q) \in \text{comp}(\text{PPI}, \text{PP}) \cap \{\text{PP}, \text{PPI}\}$, etc. Each option puts the realising occurrence pair (u, v) on the same vertical backbone through the source occurrence, giving case (a). The equality-mate variants of p or q are handled by case (b).

In every subcase, (u, v) satisfies (a) directly or admits $\equiv_{\text{EQ}}$ -mate substitutes that do, which is (b). By (A5) (Quotient-compatible pair labels), the substitution preserves lab , so the resulting Reach_R -chain witnesses $\bar{\text{lab}}([u], [v]) = R$ in the quotient. $\square$

Remark 7.2. *Lemma 7.1 is the formal counterpart of Repair R1 of GPT-5.5’s fourth review. It plugs the gap (G1): sibling-frontier mosaic positions cannot, by (M6), produce PP/PPI labels invisible to vertical reachability, so the PP/PPI universal discharge clauses (V3) and (V5) apply uniformly to every quotient R -successor. No additional non-vertical PP/PPI targets need to be considered in the truth proof.*

7.3 The standalone typed-EQ quotient theorem

Theorem 7.3 (Typed-EQ quotient). *Let $\mathcal{Q}$ be a valid finite generated split-forest certificate and $\equiv_{\text{EQ}}$ the lift of $\equiv_{\text{EQ}}^Q$ to $\Omega = \Omega(\mathcal{U}(\mathcal{Q}))$, satisfying (A1)–(A6). The quotient $\mathcal{U}(\mathcal{Q})/\equiv_{\text{EQ}}$ is a strong abstract RCC5 frame, and for every $D \in \text{cl}(C_0)$ and $u \in \Omega$,*

$$D \in \lambda(u) \iff \mathcal{U}(\mathcal{Q})/\equiv_{\text{EQ}}, [u] \models D.$$

The theorem follows from four preservation lemmas, one per property listed in GPT-5.5’s standalone-quotient request. Throughout, write $\bar{\text{lab}}([u], [v]) := \text{lab}(u, v)$, which is well-defined by (A5).

Lemma 7.4 (JEPD). *On $\mathcal{U}(\mathcal{Q})/\equiv_{\text{EQ}}$, $\bar{\text{lab}}([u], [v])$ is a unique base relation, with $\bar{\text{lab}}([u], [v]) = \text{EQ}$ iff $[u] = [v]$.*

Proof. Uniqueness is well-definedness of $\bar{\text{lab}}$, which is (A5). If $[u] = [v]$, then any representative pair (u, v') with $v' \equiv_{\text{EQ}} v$ gives $\text{lab}(u, v')$; in particular, the reflexive choice $v' = u$ gives $\text{lab}(u, u) = \text{EQ}$ by (M1). Conversely, suppose $\bar{\text{lab}}([u], [v]) = \text{EQ}$ for some representatives. Vertical comparability would forbid $\text{lab} = \text{EQ}$ (the vertical labelling reserves EQ for identical occurrences and explicit equality-port targets), so the pair is related either by occurrence identity or by an explicit equality chain in $\equiv_{\text{EQ}}^Q$. Either case forces $u \equiv_{\text{EQ}} v$, hence $[u] = [v]$. $\square$

Lemma 7.5 (Strong-EQ frame). *$\mathcal{U}(\mathcal{Q})/\equiv_{\text{EQ}}$ satisfies the abstract RCC5 frame axioms (F1)–(F3).*

Proof. (F1) is Lemma 7.4. (F2) follows from (M2) on representatives and (A5). For (F3), take any triple of classes $([u], [v], [w])$. Lemma 5.16 (patchwork) gives $\text{lab}(u, w) \in \text{comp}(\text{lab}(u, v), \text{lab}(v, w))$ on $\mathcal{U}(\mathcal{Q})$ for representatives, and (A5) lets the entries descend to $\bar{\text{lab}}$. $\square$

Lemma 7.6 (RCC5 composition). *For all $[u], [v], [w] \in \mathcal{U}(\mathcal{Q})/\equiv_{\text{EQ}}$, $\bar{\text{lab}}([u], [w]) \in \text{comp}(\bar{\text{lab}}([u], [v]), \bar{\text{lab}}([v], [w]))$.*

Proof. $\bar{\text{lab}}([u], [w]) = \text{lab}(u, w) \in \text{comp}(\text{lab}(u, v), \text{lab}(v, w)) = \text{comp}(\bar{\text{lab}}([u], [v]), \bar{\text{lab}}([v], [w]))$ by Lemma 5.16 and (A5). $\square$

Lemma 7.7 (Closure-formula truth). *For every $D \in \text{cl}(C_0)$ and every $u \in \Omega$,*

$$D \in \lambda(u) \iff \mathcal{U}(\mathcal{Q})/\equiv_{\text{EQ}}, [u] \models D.$$

Proof. By induction on $D \in \text{cl}(C_0)$. Interpret concept names by $A^{\mathcal{I}} = \{[u] : A \in \lambda(u)\}$; this is well-defined by (A3).

Atomic and Boolean cases. Direct from the definition of $A^{\mathcal{I}}$ and Hintikka completeness on closure types.

Existential $\exists R.D \in \lambda(u)$. For $R = \text{EQ}$: (V1) normalises $\exists \text{EQ}.D \in \lambda(u)$ iff $D \in \lambda(u)$, and strong-EQ semantics makes $[u] \models \exists \text{EQ}.D$ iff $[u] \models D$, so the IH closes the case. For $R \in \{\text{PP}, \text{PPI}\}$: let C be the context scheme generating the occurrence u at port (s, p) . (V4) supplies a mate port $(\hat{s}, \hat{p}) \in \text{Mate}_Q(C, s, p)$ and a target port (t, q) with $(\hat{s}, \hat{p}) \rightsquigarrow_R (t, q)$ and $D \in \lambda(t)$. By the mate-cluster construction (§6.3), C generates jointly with u a fresh mate occurrence $\hat{u}$ at port $(\hat{s}, \hat{p})$ with $u \equiv_{\text{EQ}} \hat{u}$; the strict vertical chain from $(\hat{s}, \hat{p})$ to (t, q) in Q unfolds to a strict vertical R -path from $\hat{u}$ to a fresh occurrence w of state t at port q . This is the occurrence-level (A4). By (A5), $\text{lab}([u], [w]) = \text{lab}(\hat{u}, w) = R$; by IH, $[w] \models D$; so $[u] \models \exists R.D$. Note (A2) ensures $[u] \neq [w]$. For $R \in \{\text{DR}, \text{PO}\}$: (V6) names a mosaic position q with $L(p, q) = R$ and $D \in \tau(q)$ in some mosaic M realised in the unfolding. The unfolding contains an occurrence w at q with $\text{lab}(u, w) = R$ and $D \in \lambda(w)$; (A5) descends to $\text{lab}([u], [w]) = R$, and IH gives $[w] \models D$.

Conversely, suppose $[u] \models \exists R.D$ but $\exists R.D \notin \lambda(u)$. Closure under NNF complement gives $\forall R.\bar{D} \in \lambda(u)$, where $\bar{D}$ is the NNF complement of D . For $R \in \{\text{PP}, \text{PPI}\}$, Lemma 7.1 expresses the witness pair as either a vertical Reach_R -chain from u or such a chain from an $\equiv_{\text{EQ}}$ -mate of u ; in either case (V5) applied via (A4) gives $\bar{D} \in \lambda(w)$, contradicting $D \in \lambda(w)$. For $R \in \{\text{DR}, \text{PO}\}$, (V7) directly gives $\bar{D} \in \lambda(\cdot)$ at every mosaic-supplied R -target. In all cases the contradiction forces $\exists R.D \in \lambda(u)$.

Universal $\forall R.D \in \lambda(u)$. Dual to the existential case. For $R \in \{\text{PP}, \text{PPI}\}$, Lemma 7.1 ensures that every quotient R -successor $[w]$ of $[u]$ is realised by an explicit vertical Reach_R -chain in $\mathcal{U}(\mathcal{Q})$ after $\equiv_{\text{EQ}}$ -mate substitution; (V3) and (V5) then force $D \in \lambda(w)$. For $R \in \{\text{DR}, \text{PO}\}$, (V7) covers every mosaic-supplied R -target. In every case $[w] \models D$ by IH. The converse uses NNF complement as above. $\square$

Proof of Theorem 7.3. Lemmas 7.4 and 7.5 give the strong abstract RCC5 frame. Lemma 7.6 states the composition closure explicitly (already used inside Lemma 7.5, but isolated here per GPT-5.5's request for an enumerated preservation property). Lemma 7.7 gives the closure-formula truth biconditional. $\square$

7.4 Soundness of valid certificates

The compressed lemma label `lem:eq-quotient` is retained as a synonym for the strong-EQ frame lemma so that earlier back-references remain valid.

Lemma 7.8 (Strong equality quotient; alias of Lemma 7.5). *Under (A1)–(A6), $\mathcal{U}(\mathcal{Q})/\equiv_{\text{EQ}}$ is a strong abstract RCC5 frame.*

Theorem 7.9 (Soundness of valid certificates). *If C_0 has a valid finite generated split-forest certificate, then C_0 is satisfiable in a strong abstract RCC5 frame.*

Proof. Let $\mathcal{Q}$ be valid. The validity clauses (V1)–(V10) instantiate Assumptions (A1)–(A6) on $\Omega(\mathcal{U}(\mathcal{Q}))$: (V9.a)+(V9.d) give (A1)+(A3); (V9.b) gives (A2); (V4) together with the port-level mate-cluster construction of §6.3 (whereby every generated occurrence u of source state s at port p simultaneously generates fresh mate occurrences for every $(\hat{s}, \hat{p}) \in \text{Mate}_Q(C, s, p)$) gives (A4); (V9.c) gives (A5); (V9.c) on mosaic positions plus (SC3) gives (A6). By Theorem 7.3, $\mathcal{U}(\mathcal{Q})/\equiv_{\text{EQ}}$ is a strong abstract RCC5 frame and the closure-formula truth biconditional holds at every occurrence. Since $C_0 \in \lambda(s_0)$ and u_0 is the distinguished initial occurrence of s_0 , the class $[u_0]$ satisfies C_0 . $\square$

8 Completeness

Completeness has three steps: thinning, split cover extraction, and finite blocking. The first two were Lemmas 3.7 and 4.3. It remains to show that the split presentation can be represented by a finite valid certificate.

8.1 Complete finite profiles

The profile used for blocking is intentionally richer than a closure type. It contains every datum needed to replace one generated subtree by another without changing satisfaction of closure formulas or mosaic compatibility at the boundary.

Definition 8.1 (Complete boundary profile). *For an occurrence u in a generated split presentation, its complete boundary profile $\Pi(u)$ consists of:*

- (B1) *the closure type $\lambda(u)$;*
- (B2) *the finite list of selected existential obligations active at u ;*
- (B3) *for each PP or PPI obligation, whether it is discharged in the current finite context, through which equality mate it is discharged, or through which vertical boundary it is passed;*
- (B4) *for each DR or PO obligation, the actual finite side mosaic position witnessing it;*
- (B5) *the equality-port pattern of all split copies of the semantic element that occur in the current boundary context;*
- (B6) *the relation matrix and type labels of every boundary position of the current context;*
- (B7) *the sets of closure formulas true at all and at some strict vertical ancestors and descendants reachable through each boundary port.*

Only finitely many complete boundary profiles exist for a fixed C_0 , because $\text{cl}(C_0)$, the role set, the context width, the equality-port alphabet, and the RCC5 label set are finite. The context width can be fixed effectively: one source position, one selected witness position for each existential closure formula, equality-mate positions needed to realize the selected containment parents of those witnesses, and the finitely many positions required to check all pair and triple interfaces. A crude bound exponential in $n = |\text{cl}(C_0)|$ is sufficient.

Lemma 8.2 (Substitution lemma). *If two boundary occurrences u and v in a generated split presentation have the same complete boundary profile, then the generated component above, below, or sideways from u may be replaced by a fresh copy of the corresponding component from v , preserving all closure types, all explicit equality-port interfaces, all PP/PPI reachability summaries, and all mosaic compatibility conditions at the boundary.*

Proof. Every possible interaction between the replaced component and the outside occurs through a boundary port or a mosaic boundary position. Equality-port data agree by (B5), relation matrices and type labels agree by (B6), and vertical reachability summaries agree by (B7). Existential witnesses either remain inside the copied component, are supplied by the same boundary port, or are supplied by an explicitly named side mosaic, all of which are part of the profile. Universal restrictions depend only on the relation label and the target type for every reachable or mosaic-visible target; these are also part of the profile. Finally, every triple crossing the boundary is represented by the same abstract triple mosaic before and after replacement. Hence no closure formula or RCC5 composition condition can distinguish the original component from the copied one. $\square$

8.2 Finite-index blocking without automata

We now prove regularity directly, using request-fulfilled cycles rather than automata. The argument has three ingredients: (i) a finite alphabet of *extended profiles* that records both the complete boundary profile and the set of still-pending vertical existentials, (ii) a checkpoint construction that produces infinitely many positions where every request pending at the preceding checkpoint has already been fulfilled, and (iii) a pigeonhole that selects two equi-profile checkpoints whose intervening segment is guaranteed to contain a strict later fulfilling position for every R -request inside it.

For a vertical direction $R \in \{\text{PP}, \text{PPI}\}$, an R -request at an occurrence u is an existential formula $\exists R.D \in \lambda(u)$ not already witnessed inside the current finite context. A segment of a vertical path from u_i to u_j is *request-fulfilled* if every R -request that is pending at some position u_k in the segment ($i \leq k \leq j$) is discharged by a strict later position $u_{k'}$ in the *same* segment, $k < k' \leq j$, whose type contains the witness formula. This is strictly stronger than the deferral reading: a request is never merely passed out at u_j and left to be re-tried in some future lap; the segment itself supplies a strict witness, and pumping the segment forever turns a request-fulfilled cycle into actual fulfillment of every existential along the resulting path. In particular, a cycle that carries a request $\exists R.D$ but does not contain a strict later position with $D \in \lambda$ is *not* request-fulfilled and is rejected by the validity check.

Definition 8.3 (Request status and extended profile). *For an occurrence u_k on an infinite vertical path $u_0, u_1, \dots$, the pending set $\text{Pend}(u_k) \subseteq \{\exists R.D \in \text{cl}(C_0) : R \in \{\text{PP}, \text{PPI}\}\}$ is the set of vertical existentials generated at some occurrence $u_{k'}$ with $k' \leq k$ and not yet witnessed by any $u_{k''}$ with $k' \leq k'' \leq k$. The extended profile of u_k is*

$$\hat{\Pi}(u_k) := (\Pi(u_k), \text{Pend}(u_k)).$$

The pending set ranges over a finite alphabet: it is a subset of the at-most- n vertical existentials in $\text{cl}(C_0)$, so $|\text{Pend}| \leq 2^n$. Combined with the finitely many complete boundary profiles, the extended-profile alphabet $\hat{\Sigma}$ is finite of size $\leq |\Sigma| \cdot 2^n$.

Lemma 8.4 (Checkpoint structure). *Suppose every transitive existential request on $u_0, u_1, \dots$ is eventually fulfilled. For each $k \geq 0$ there is a least position*

$$\text{chk}(k) := \min\{m > k : \text{every } \exists R.D \in \text{Pend}(u_k) \text{ is fulfilled at some } u_{k'}, k \leq k' \leq m\}.$$

Iterating $k_0 := 0, k_{n+1} := \text{chk}(k_n) + 1$ produces an infinite increasing sequence of checkpoint positions.

Proof. For each $\exists R.D \in \text{Pend}(u_k)$, by hypothesis there is some $m_{\exists R.D} > k$ at which it is fulfilled. Take $\text{chk}(k)$ to be the maximum of these (at most n) values: it exists since $\text{Pend}(u_k)$ is finite. Iteration is well-defined because $k_{n+1} > k_n$ for every n . $\square$

Lemma 8.5 (Request-fulfilled repetition lemma). *Let*

$$u_0, u_1, u_2, \dots$$

be an infinite generated vertical path in a split presentation, and suppose every transitive existential request on the path is eventually fulfilled in the presentation. Then there are $i < j$ such that $\Pi(u_i) = \Pi(u_j)$ and the segment $u_i, \dots, u_j$ is request-fulfilled: for every transitive existential request $\exists R.D$ that is pending at some position u_k with $i \leq k \leq j$, there is a strict later position $u_{k'}$ within the same segment, $k < k' \leq j$, with $D \in \lambda(u_{k'})$. Consequently the pumped path

$$u_0, \dots, u_{i-1}, (u_i, \dots, u_{j-1})^\omega$$

built from fresh occurrences of the segment satisfies the strong fulfillment property:

every PP/PPI existential request generated at any occurrence of the pumped path is fulfilled at a strict later occurrence on the same path, within bounded lookahead $\leq j - i$.

All PP/PPI universal closure formulas are preserved at every occurrence of the pumped path.

Proof. Pigeonhole. By Lemma 8.4 the sequence of checkpoints $u_{k_0}, u_{k_1}, \dots$ is infinite. Each u_{k_n} has an extended profile $\hat{\Pi}(u_{k_n}) \in \hat{\Sigma}$. Since $\hat{\Sigma}$ is finite, the infinite sequence $\hat{\Pi}(u_{k_n})$ takes some value at least twice: there are $n_1 < n_2$ with $\hat{\Pi}(u_{k_{n_1}}) = \hat{\Pi}(u_{k_{n_2}})$. Set $i := k_{n_1}$, $j := k_{n_2}$. In particular $\Pi(u_i) = \Pi(u_j)$ and $\text{Pend}(u_i) = \text{Pend}(u_j)$.

Strict cyclic fulfillment. We show two facts.

(F1) *The segment supplies a witness label for every pending formula.* Let $\exists R.D \in \text{Pend}(u_j)$. By the pigeonhole choice of i, j , $\text{Pend}(u_j) = \text{Pend}(u_i)$, so $\exists R.D \in \text{Pend}(u_i)$. By the checkpoint definition, $\exists R.D$ is fulfilled in the original path at some u_m with $i \leq m \leq \text{chk}(i) \leq j - 1 < j$, so $D \in \lambda(u_m)$ with $i \leq m < j$. Hence every formula in $\text{Pend}(u_j) = \text{Pend}(u_i)$ has a witness label occurring somewhere in the finite segment $[u_i, u_{j-1}]$.

(F2) *The pumped path realizes a strict later witness for every existential, within bounded lookahead.* Consider any occurrence $u_k^{(\ell)}$ of the pumped path ($i \leq k < j$, $\ell \geq 0$) and any $\exists R.D \in \lambda(u_k)$ that is generated at $u_k^{(\ell)}$. Two cases.

(F2a) If $\exists R.D$ is fulfilled in the original path at some $u_{k'}$ with $k < k' \leq j - 1$, the pumped path contains the fresh occurrence $u_{k'}^{(\ell)}$ with $D \in \lambda(u_{k'}^{(\ell)}) = \lambda(u_{k'})$, strictly later than $u_k^{(\ell)}$ in the same lap. Lookahead: $k' - k \leq j - 1 - k < j - i$.

(F2b) Otherwise $\exists R.D$ remains in $\text{Pend}(u_j) = \text{Pend}(u_i)$. By (F1), $D \in \lambda(u_m)$ for some $i \leq m \leq \text{chk}(i) \leq j - 1$. In the pumped path, the occurrence $u_m^{(\ell+1)}$ in the next lap satisfies $D \in \lambda(u_m^{(\ell+1)})$ and lies strictly later than $u_k^{(\ell)}$. Lookahead: $(j - k) + (m - i) + 1 \leq 2(j - i)$.

Together (F1) and (F2) establish that the segment $u_i, \dots, u_{j-1}$ is *request-fulfilled*: every transitive existential generated at any occurrence of the pumped path has a strict later witness in the pumped path, within bounded lookahead $\leq 2(j - i)$. No request is merely deferred to a future lap; the finite segment itself supplies the witness label, and pumping guarantees that the witness label is encountered as a strict later occurrence.

Pumping. Replace the tail $u_j, u_{j+1}, \dots$ by infinitely many fresh copies $u_i^{(1)}, \dots, u_j^{(1)}, u_i^{(2)}, \dots, u_j^{(2)}, \dots$ of the segment $[u_i, u_j]$, where each $u_k^{(\ell)}$ is a fresh occurrence of the same profile as u_k . By $\Pi(u_i) = \Pi(u_j)$, the splice is profile-consistent; by the substitution lemma (Lemma 8.2), local closure formulas and mosaic compatibility at the splice point are preserved.

Preservation of universals. Every $\forall R.D$ formula at $u_k^{(\ell)}$ depends only on the relations and target types visible from $u_k^{(\ell)}$. By substitution the visible neighbourhood of $u_k^{(\ell)}$ is the same as the visible neighbourhood of u_k , so $\forall R.D$ truth is inherited from the original path.

Preservation of existentials. This is exactly (F2) above: every PP/PPI existential generated at any pumped-path occurrence has a strict later witness within bounded lookahead $\leq 2(j - i)$. No indefinite deferral is admitted, and pumping a request-fulfilled segment yields an infinite path satisfying every existential.

Thus every closure formula at every fresh occurrence of the pumped path holds, completing the proof. $\square$

Lemma 8.6 (Finite regularization lemma). *Every generated split presentation satisfying all closure formulas and support-closed mosaic conditions has a finite regular split-forest certificate satisfying Definition 6.6.*

Proof. Label every boundary occurrence by its complete profile. There are finitely many labels. For finite branches, keep one representative context for each complete profile encountered before termination. For an infinite vertical ray, apply Lemma 8.5 and replace the tail by the request-fulfilled repeated segment. This produces a finite stem and a finite blocking cycle. Because repeated laps are fresh, no semantic equality is introduced by this operation.

For branching, perform the same replacement independently at each boundary profile. The substitution lemma guarantees that replacing a component by another component with the same complete profile preserves all interactions through the boundary. Since each profile has only finitely many selected existential obligations and side contexts, only finitely many representative outgoing contexts are needed. Collect these representatives into the finite state set S , record the parent map and child multisets, and record the exact context and mosaic schemes witnessed by the representatives.

Equality data are recorded only as explicit port links inherited from equality of split copies in the original presentation. If two occurrences are merely blocked copies of the same profile, no equality link is recorded. Since the original presentation quotients to a strong frame, no explicit equality link connects strict vertical comparables, and equality mates have identical types and identical external relation behaviour. Hence the typed equality congruence clauses hold.

The reachability summaries in the profiles are copied from the actual representative components and then recomputed on the finite quotient. Request-closed cycles guarantee that recomputation agrees with the copied summaries: a target available after another lap is represented by reachability around the blocking cycle, while distinct laps remain fresh occurrences. Side witnesses and triple support are preserved by the collected support-closed mosaic family. Therefore all validity conditions (V1)–(V10) hold. $\square$

Theorem 8.7 (Completeness of certificates). *If C_0 is satisfiable, then C_0 has a valid finite generated split-forest certificate.*

Proof. Let $\mathcal{I}, a_0 \models C_0$. By Lemma 3.7, replace $\mathcal{I}$ by a witness-generated induced submodel $\mathcal{J}, a_0$ preserving all closure formulas. By Lemma 4.3, construct a generated split presentation of $\mathcal{J}$. This presentation satisfies all closure formulas by pullback from $\mathcal{J}$, and its finite side interfaces are legal because their labels are inherited from the strong RCC5 frame $\mathcal{J}$.

Apply the finite regularization lemma to obtain a finite certificate. All validity clauses are inherited from the presentation and preserved by substitution. The distinguished profile contains C_0 because the distinguished occurrence has origin a_0 and $a_0 \models C_0$. $\square$

9 Decision procedure

This section makes the decision procedure effective by giving (i) a compact formal grammar for certificates, (ii) explicit polynomial bounds on each component alphabet, and (iii) an enumeration bound sufficient to make the satisfiability test an exhaustive finite search. The bound is coarse; computability is what is needed for decidability.

9.1 Grammar of certificate components

Fix the input C_0 and set $n := |\text{cl}(C_0)| \leq 2|\text{sub}(C_0)|$. A finite generated split-forest certificate is the formal expression of the following abstract grammar (writing $\sqcup$ for choice and $\langle \cdot, \dots, \cdot \rangle$ for tuples):

$$\begin{aligned}
\text{Certificate } \mathcal{Q} &::= \langle S, s_0, \lambda, \text{par}_Q, \text{Ch}_Q, \text{Ctx}_Q, \text{Mos}_Q, E_Q, \text{wit}_Q \rangle \\
\text{StateSet } S &::= \emptyset \sqcup \{\sigma\} \cup S, \quad \sigma \in \Sigma \\
\text{Profile } \sigma &::= \langle \tau, \kappa, \mu, \eta \rangle \\
\text{Type } \tau &\subseteq \text{cl}(C_0), \quad \tau \in \text{Typ}(C_0) \\
\text{ContextColour } \kappa &::= \langle \kappa_{\uparrow}, \kappa_{\downarrow}, \kappa_{\text{side}} \rangle \\
\text{SideMenu } \mu &::= \{ \exists R.D \mapsto (M_i, q_i) : \exists R.D \in \tau, R \in \{\text{DR}, \text{PO}\} \} \\
\text{InterfaceSummary } \eta &::= \{ (\pi, R) \mapsto \text{Reach}_R(\pi) \subseteq \text{cl}(C_0) : \pi \in \text{Port}(\sigma), R \in \{\text{PP}, \text{PPI}\} \} \\
\text{ParentMap } \text{par}_Q &::= S \rightarrow S \\
\text{ChildMultiset } \text{Ch}_Q(s) &\subseteq_{\text{mset}} S \\
\text{Context } \text{Ctx}_Q &::= \{ s \mapsto \text{context scheme} \} \\
\text{Mosaic } M &::= \langle P, \tau_M, L_M \rangle \\
\text{PositionSet } P &\subseteq \{1, \dots, m(n)\} \\
\text{Edge } L_M(p, q) &\in \mathbf{B} \\
\text{EqPort } E_Q &::= \{ (\sigma_i, p_i) \sim (\sigma_j, p_j) \} \\
\text{Witness } \text{wit}_Q &::= \{ \exists R.D \mapsto \text{mechanism} \}
\end{aligned}$$

Each terminal ranges over a finite alphabet determined by n :

- $\text{cl}(C_0)$: $\leq 2|\text{sub}(C_0)| \leq 2n$.
- $\text{Typ}(C_0)$: $\leq 2^n$.
- $\text{Port}(\sigma)$: bounded by $n + 1$ (one port per PP/PPI obligation, plus the identity port).
- Context colours κ : three polynomial slots, each carrying one of finitely many context scheme names.
- Mos_Q : mosaics with position count $\leq m(n)$ where $m(n)$ is the crude width fixed below (one source, one position per existential formula in τ , one equality-mate boundary position per containment witness, and one position per realised triple of such positions).

9.2 Explicit component bounds

Fix a polynomial bound $m(n) = c \cdot n^k$ for the mosaic position count, with constants c, k determined by the context schemes (the choice $c = 8$, $k = 2$ already suffices: each existential $\exists R.D \in \tau$ contributes one source and one witness position, and triple closure quadruples this). Then:

- **Closure types.** $|\text{Typ}(C_0)| \leq 2^n$.
- **Equality-mate interface ports.** $|\text{Port}(\sigma)| \leq n + 1$, so the port alphabet across profiles has size $\leq |\Sigma| \cdot (n + 1)$.
- **Selected-parent and selected-child slots.** par_Q is a partial function $S \rightarrow S$, giving $|S|^{|S|+1}$ choices; Ch_Q is a multiset on S with capacity bounded by the existentials in $\lambda(s)$, so $|\text{Ch}_Q(s)| \leq n$ and the number of child assignments per state is $\leq n \cdot |S|^n$.

- **Request summaries.** η records, per port and per vertical direction, a subset of $\text{cl}(C_0)$: $(n+1) \cdot 2 \cdot 2^{2n} = O(n \cdot 2^{2n})$ bits per profile.
- **Sibling/support mosaics.** A mosaic is a function $\tau_M : P \rightarrow \text{Typ}(C_0)$ together with $L_M : P^2 \rightarrow B$: $\leq (2^n)^{m(n)} \cdot 5^{m(n)^2} \leq 2^{p_1(n)}$ for $p_1(n) := n \cdot m(n) + m(n)^2 \cdot \log_2 5$. Mos_Q is a subset: $\leq 2^{2^{p_1(n)}}$ families. Maximum arity is $m(n)$.
- **Finite graph size after blocking.** By Lemma 8.6 the state set S has $|S| \leq N(C_0)$, the number of complete boundary profiles, and $N(C_0) \leq |\text{Typ}(C_0)| \cdot (\text{context colour count}) \cdot (\text{side menu count}) \cdot (\text{interface count})$. Each factor is at most $2^{p_2(n)}$ for an explicit polynomial p_2 , so $|\Sigma| \leq 2^{p(n)}$ and $|S| \leq 2^{p(n)}$.

Collecting these,

$$p(n) := n + 2 \cdot m(n) \cdot \log_2 5 + 2n \cdot (n+1) + \text{const}$$

is a polynomial of degree $\max(k+1, 2k)$ in n .

9.3 Enumeration bound

Lemma 9.1 (Certificate enumeration bound). *There is an explicit polynomial $p(n)$ such that the number of finite generated split-forest certificates for C_0 with state count $\leq B(C_0) := 2^{2^{p(n)}}$ and context width $\leq B(C_0)$ is at most $2^{2^{q(n)}}$ for some polynomial $q(n)$. Each certificate's validity (V1)–(V10) is decidable in time polynomial in its description size.*

Proof. By the grammar of §9.1, a certificate is a finite tuple over alphabets of size $\leq 2^{p(n)}$ (states, types, contexts, mosaic positions, port symbols, witness mechanisms) with at most $B(C_0) = 2^{2^{p(n)}}$ components. The total number of certificates of size $\leq B(C_0)$ is then bounded by

$$(2^{p(n)})^{B(C_0)} \cdot (\text{combinatorial overhead}) \leq 2^{p(n) \cdot 2^{2^{p(n)}}} \leq 2^{2^{q(n)}}$$

for some polynomial q .

Each validity clause (V1)–(V10) is a finite-quantifier predicate over the certificate's components: (V1)–(V3) are local type/context checks; (V4)–(V5) are graph-reachability checks on S (linear in $|S|$ per query); (V6)–(V8) are mosaic-position membership and triple checks (polynomial in mosaic count); (V9) is a congruence check on the explicit-port relation; (V10) is exact summary computation by reachability. All checks run in time polynomial in $|Q|$, which is polynomial in $B(C_0)$. $\square$

The compressed label `lem:bound` is retained as a synonym of Lemma 9.1 for back-reference compatibility.

Lemma 9.2 (Effective finite search bound; alias of Lemma 9.1). *For each input C_0 there is a computable number $B(C_0)$ such that if C_0 has a valid finite generated split-forest certificate, then it has one with at most $B(C_0)$ states and with context width at most $B(C_0)$.*

9.4 Decidability

Theorem 9.3 (Decidability). *Concept satisfiability for $\mathcal{ALCI}_{\text{RCC5}}$ over strong abstract RCC5 frames is decidable.*

Proof. Given C_0 , compute $\text{cl}(C_0)$ and the bound $B(C_0)$. Enumerate all finite tuples of the form in Definition 6.3 with at most $B(C_0)$ profile states and context width at most $B(C_0)$. By Lemma 9.1 this enumeration is finite and each tuple’s validity (V1)–(V10) is decidable in polynomial time.

If a valid certificate is found, accept; by Theorem 7.9 (soundness), C_0 is satisfiable. If no valid certificate exists below the bound, reject; by Theorem 8.7 (completeness) and Lemma 9.2, every satisfiable C_0 would have appeared in the finite search. Hence the algorithm halts with the correct answer. $\square$

10 How the repaired proof handles the known stress cases

10.1 Infinite upward chains

The formula

$$C_{\uparrow} = \exists \text{PP}.\top \sqcap \forall \text{PP}.\exists \text{PP}.\top$$

is represented by a certificate with a blocking parent self-loop on a profile containing $C_{\uparrow}$. The unfolding is

$$u_0 \text{PP} u_1 \text{PP} u_2 \text{PP} \dots,$$

where each u_i is fresh. The existential $\exists \text{PP}.\top$ at u_i is witnessed by u_{i+1} . No semantic equality is introduced by the self-loop.

10.2 Several incomparable upward witnesses

Consider a node requiring two incompatible superpart witnesses:

$$\exists \text{PP}.A \sqcap \exists \text{PP}.B$$

together with restrictions forcing the A - and B -witnesses to be incomparable. A single occurrence has only one parent chain, so the split forest creates two equality-mate occurrences of the same semantic element. One mate is placed below an A -superpart and one below a B -superpart. In the quotient these mates become one element. Existential discharge is equality-aware: an occurrence may satisfy its $\exists \text{PP}.A$ obligation through the parent chain of one mate and its $\exists \text{PP}.B$ obligation through the parent chain of another mate. Typed equality congruence ensures that all universal restrictions and external RCC5 labels remain invariant under the collapse.

10.3 DR/PO witnesses

A DR or PO existential is never discharged by a raw pair table entry detached from a concrete occurrence. It is discharged only by a named side position in a legal mosaic containing the source, the witness, and the relevant boundary positions. Thus relation support and witness support are joint data, and the proof does not combine witnesses from incompatible contexts.

11 What remains of the original split-forest architecture

The repaired proof keeps the original architecture in the following precise form.

Original component	Repaired status
Tree-shaped containment backbone	Survives as a generated witness-cover tree, not as the Hasse diagram of an arbitrary starting model.
Split copies	Survive; they are essential for elements with several selected containment parents.
Sibling interfaces	Survive as support-closed mosaics rather than raw pair tables.
DR/PO sibling simplification	Survives for ordinary non-core sibling frontier pairs after containment, core copies, and equality interfaces are accounted for.
Typed EQ quotient	Survives as an explicit congruence; it is never inferred from profile equality or blocking.
Finite quotient	Survives as a finite regular certificate whose unfolding may be infinite.
Blocking	Survives as finite-index repetition with fresh laps and request-fulfilled cycles.

The proof therefore does not abandon the split-forest idea. It makes explicit the two preliminary reductions needed to reach the intended model class: witness thinning and generated cover presentation. Once those are in place, the proof operates on the sparse generated structures originally envisioned, while the finite certificate records enough equality, reachability, and mosaic information to make the construction sound.